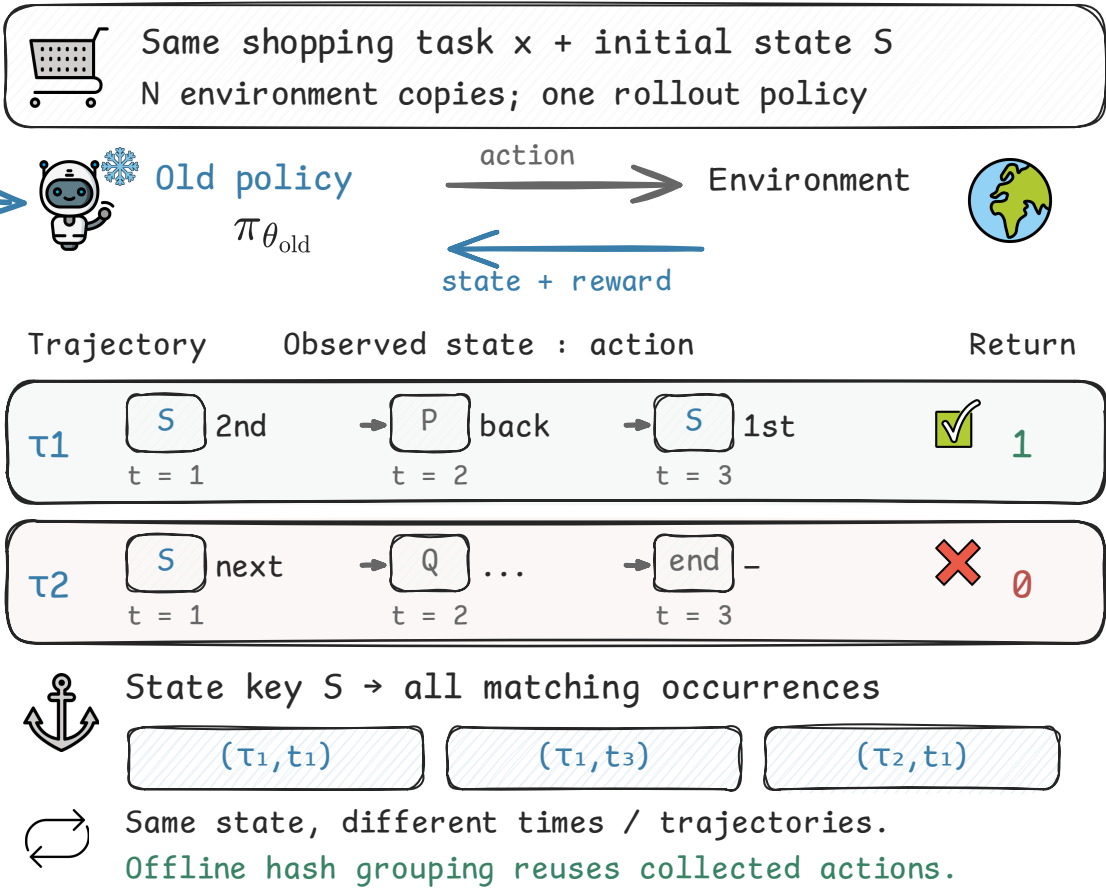


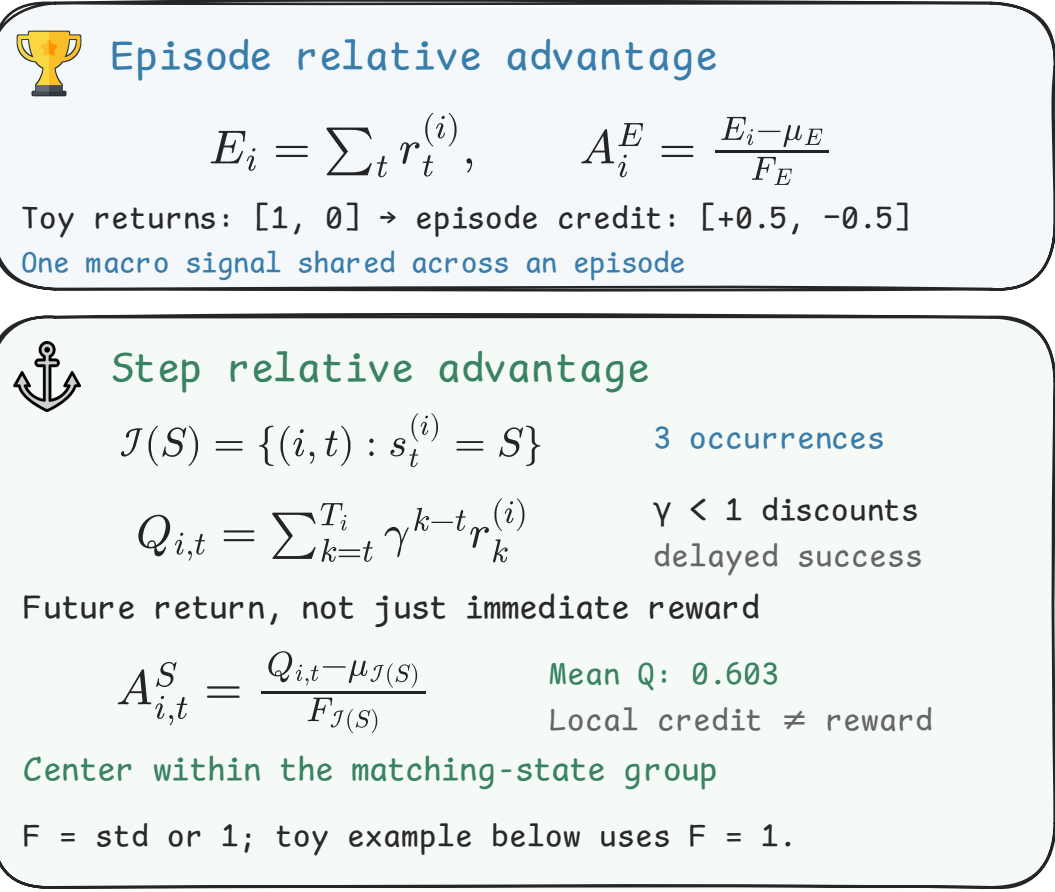
GiGPO

Group-in-group credit assignment for multi-turn agents

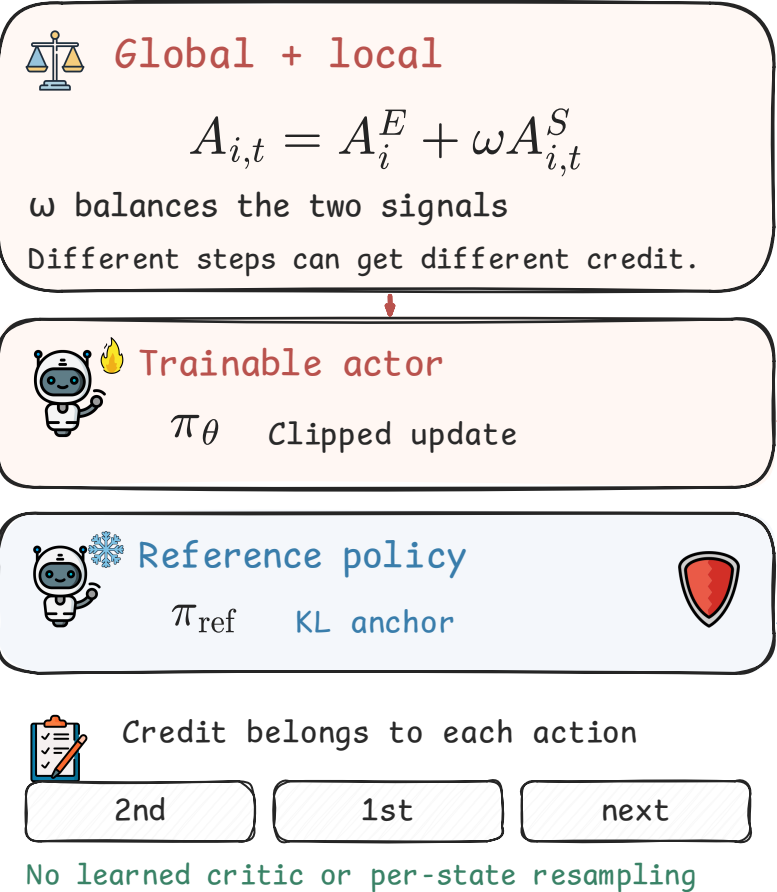
Collect a shared trajectory group



Episode & anchor-state comparisons



Fuse credit & update



Clipped action objective + reference KL

$$\hat{J} = \frac{1}{NT} \sum_{i,t} \min(\rho_{i,t} A_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_{i,t}) - \beta D_{\text{KL}}(\pi_{\theta} \| \pi_{\text{ref}})$$

Worked anchor group S $\gamma = 0.9, \omega = 1, F = 1$

Occurrence / action	Q	Episode	Step	Combined
τ_1, t_1 : 2nd item	0.81	+0.50	+0.21	+0.71
τ_1, t_3 : 1st item	1.00	+0.50	+0.40	+0.90
τ_2, t_1 : next page	0.00	-0.50	-0.60	-1.10

